

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AP40016
Subject Title	Laser Processing and Applications
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	The course introduces various advanced laser technologies and its applications including lasers and 3D printing. Students can also understand the new concepts of smart factory by implementing the predictive maintenance, smart sensors, AI, large-scale machine-to-machine communication (M2M) and the internet of things (IoT) for increased automation, resources optimization and self-monitoring.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) Describe the general operational principles of CW, Q-switched, mode locked; understand the concepts of lineshape broadening, three level and four level laser systems; (b) understand the difference between spontaneous and stimulated emission, laser features; population inversion and the stability of a laser resonator, (c) describe the general operational principle of laser applications in 3D printing technology, medical physics, communication, pulsed laser deposition, military, manufacturing etc.; understand the laser and matter interaction; laser safety, (d) understand the importance and applications of AI for manufacturing; smart factory involving predictive maintenance, smart sensors, IoT, M2M AL for manufacturing; (e) apply the knowledge to solve some real life problems.
Subject Synopsis/ Indicative Syllabus	Laser principles of laser: three level and four level lasers system, wavelength line-shape broadening, coherence of radiation, stimulated emission, laser gain media, techniques of pumping, laser resonator, laser mode, Q-switched and mode locked lasers. Study design of various types of laser systems: ultra-fast lasers, gas discharge lasers, diode pumped solid state lasers, semiconductor lasers, etc. Laser characterization: beam quality, pulse width, repetition rates, intensity, wavelengths FWHM, mode matching, laser safety, laser light coupling. Interaction of light with matter: molecular energy levels, absorption, stimulated and spontaneous emission, scattering. Examples of laser applications: laser for material processing, laser for medical applications, laser for scientific applications, e.g. absorption spectroscopy; laser tweezers, etc. Military: laser guided missile, target designator, pulsed laser deposition, 3D printing etc. 3D printing ad AI technology: General working principles, features of different categories of printing technologies, printing materials, multi-material 3D printing, colour 3D printing etc. Applications of 3D printing, health and safety issues, possible impact to the industry and society. The implementation of predictive maintenance, smart sensors, and the internet of things (IoT) and AI to revolutionize the future manufacturing.

Teaching/Learning Methodology	Lecture: The course contents will be introduced and discussed in lectures. An active participation in these learning activities can significantly enhance students' conceptual understanding, knowledge acquisition and problem-solving skills. Student-centered Tutorial: Tutorial classes allow students to ask questions related to abstract and difficult learning content, followed by in-depth explanation and discussion, resulting in fostering their skills in analytical power, concept acquisition, critical thinking, problem-solving and life-long learning.																																						
Assessment Methods in Alignment with Intended Learning Outcomes	<table border="1"> <thead> <tr> <th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr> <tr> <th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr> </thead> <tbody> <tr> <td>1. Continuous assessment</td><td>40</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr> <tr> <td>2. Examination</td><td>60</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr> <tr> <td>Total</td><td>100</td><td colspan="5"></td></tr> </tbody> </table> Continuous assessment: The continuous assessment includes assignments, quizzes and test(s) which aim at checking the progress of students study throughout the course, assisting them in fulfilling the learning outcomes. Assignments in general include end-of-chapter problems, which are used to reinforce and assess the concepts and skills acquired by the students; and to let them know the level of understanding that they are expected to reach. At least one test would be administered during the course of the subject as a means of timely checking of learning progress by referring to the intended outcomes, and as means of checking how effective the students digest and consolidate the materials taught in the class. Examination: This is a major assessment component of the subject. It would be a closed-book examination.						Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					a	b	c	d	e	1. Continuous assessment	40	✓	✓	✓	✓	✓	2. Examination	60	✓	✓	✓	✓	✓	Total	100					
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1. Continuous assessment	40	✓	✓	✓	✓	✓																																	
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Student Study Effort Expected	Class contact:																																						
	▪ Lecture & Tutorial					39 Hrs.																																	
	Other student study effort:																																						
	▪ Self-study					81 Hrs.																																	
	Total student study effort					120 Hrs.																																	
Reading List and References	1. Walter Koechner, 'Solid-State Laser Engineering', 6th Edition, Springer Berlin / Heidelberg, 2010. 2. Gennady G. Gladush, 'Physics of Laser Materials Processing: Theory and Experiment', 1st Edition, Springer, 2011. 3. R. Henderson and K. Schulmeister, 'Laser Safety: For Users and Manufacturers of Laser Equipment', 2nd Edition, Taylor & Francis, 2011. 4. Aranda, Sean, Feeney, David, 3D Printing Failures: 2020 Edition: How to Diagnose and Repair ALL Desktop 3D Printing Issues, Independently published, 2020. 5. O. Perez, S. Saucedo, J. Cruz, "Manufacturing 4.0: The Use of Emergent Technologies in Manufacturing", Palibrio, 2018.																																						